

LLM Architecture and Planning Systems



AI Tools and Prompt Engineering for PMs



Learning Objectives

By the end of this lesson, you will be able to:

- 🔗 Identify where and how AI tools and LLMs can enhance key decisions across the product management lifecycle
- 🔗 Construct high-impact prompts that generate accurate user stories, feature copy, and summaries
- 🔗 Diagnose common prompt failure patterns and revise them to improve output relevance and reliability
- 🔗 Apply Chain-of-Thought and ReAct prompting methods to structure reasoning and automate multi-step tasks
- 🔗 Select the right AI tools for tasks such as writing, summarization, UI prototyping, and project coordination



Engage and Think



Product managers (PMs) today are expected to move faster, make smarter decisions, and navigate increasing complexity. AI is no longer experimental; it is actively transforming how teams research, prioritize, and build.

As AI becomes more embedded in product work, the key question isn't just what it can do, but how you can leverage it to work better.

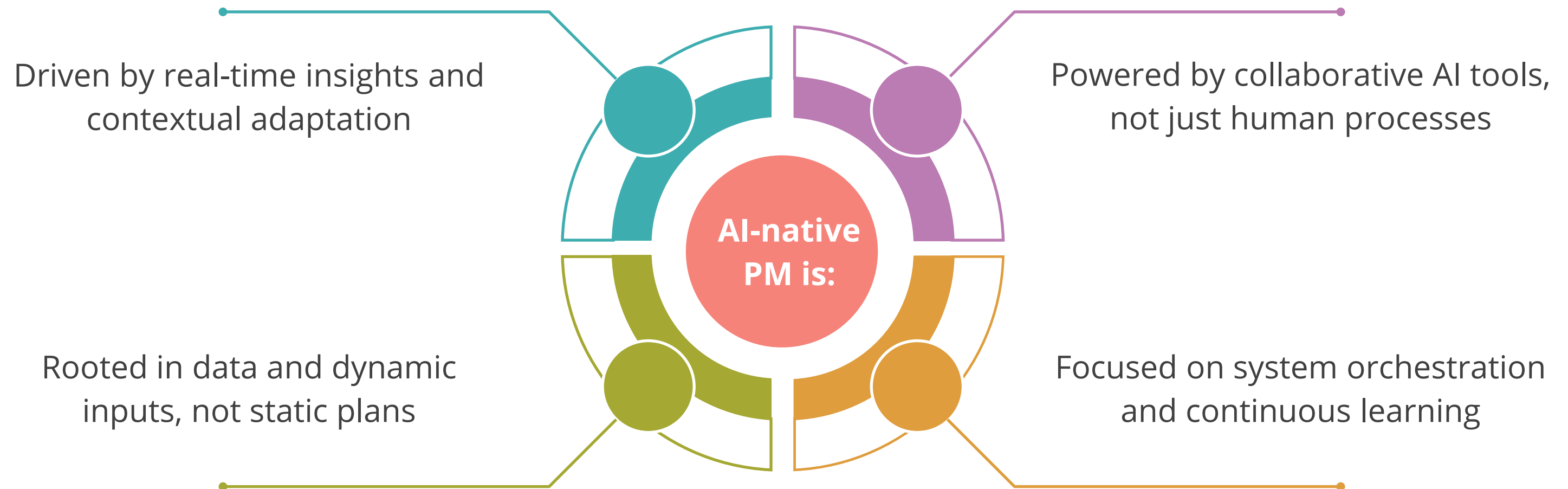
Where in your product lifecycle could AI deliver real value, not by replacing your role, but by amplifying how you think, decide, and deliver?



The AI Shift in Product Management

AI-Native Product Management (PM)

AI is transforming product management by shifting how PMs make decisions, collaborate, and build for dynamic, data-driven environments.



Shift from Traditional to AI-Native PM Practices

The responsibilities of product managers are shifting from planning and process to orchestration and speed.



Traditional PMs:

- Deliver features based on fixed requirements, design thinking, and testing
- Measure success by usage metrics and timely delivery



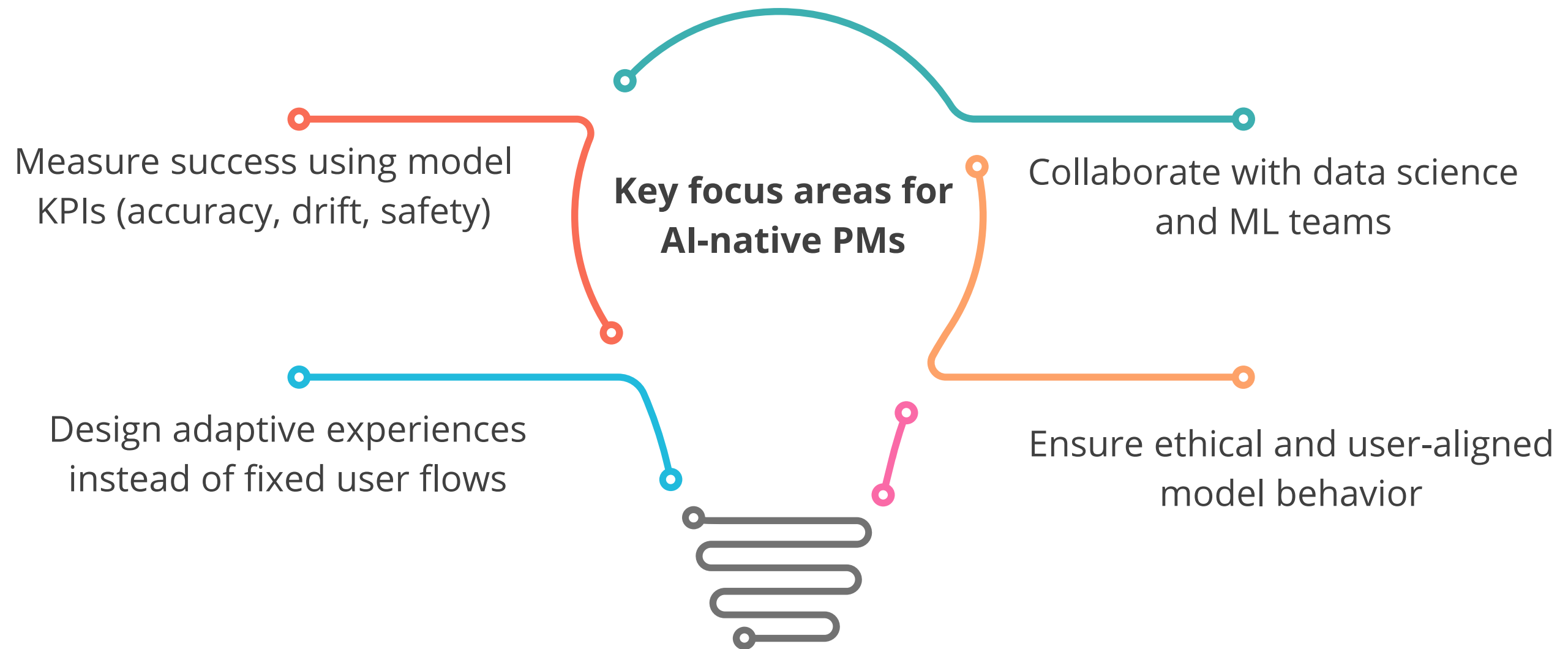
AI-native PMs:

- Build intelligent systems using data, models, and AI tools for dynamic outcomes
- Measure success by accuracy, adaptability, and learning over time

The shift from traditional to AI-native PMs is about evolving from fixed plans to adaptive systems that continuously improve and respond to real-world feedback.

Adapting PM Roles in the Era of AI-Native Apps

AI-native apps demand product managers who drive outcomes through models, metrics, and multidisciplinary collaboration.

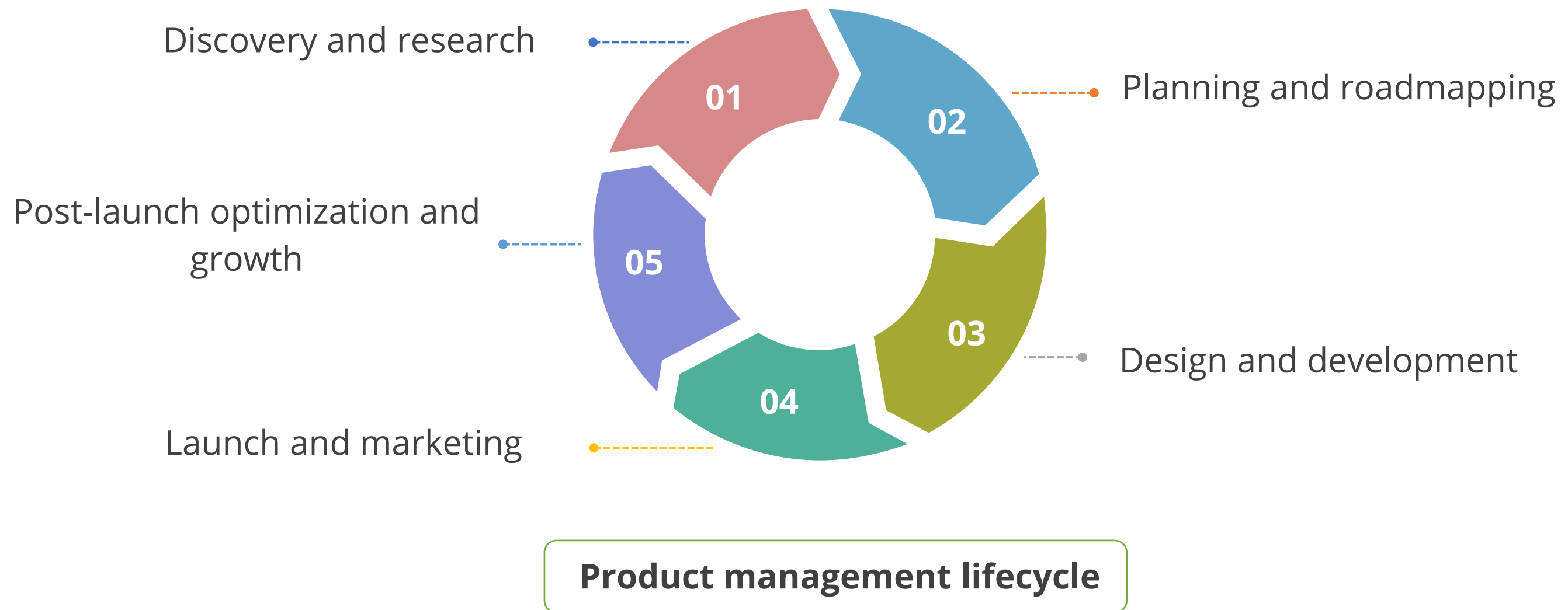




AI Across the Product Management Lifecycle

Product Management Lifecycle

AI is profoundly transforming each phase of product management, enabling smarter decisions, faster execution, and deeper user insights throughout the lifecycle.



AI in Product Discovery and Research

AI accelerates discovery by transforming how product managers gather market data and extract user insights.

Market research and analysis

Analyzes large datasets to uncover unmet needs, market gaps, and innovation opportunities



Customer feedback and insights

Analyzes behavior patterns, feedback, and logs to identify user needs and improvement areas



AI in Product Planning and Roadmapping

AI brings structure and foresight to planning by helping product managers prioritize, forecast, and align with business goals.

Backlog prioritization

Analyzes feedback and capacity to rank backlog items by impact and effort



Forecasting and risk management

Uses predictive analytics to estimate timelines and flag risks or dependencies



Goal setting and KPIs

Defines realistic goals and KPIs to align with performance and market trends

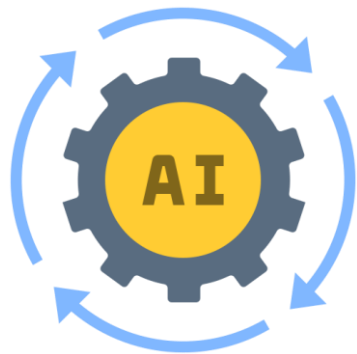


AI in Product Design and Development

AI accelerates product design by automating creative tasks, simulating performance, and ensuring quality from the start.

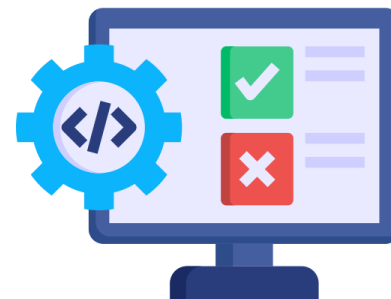
Automated design

Generates design variations to optimize aesthetics, performance, and feasibility



Prototyping and testing

Simulates prototypes and tests virtually to speed up iteration and detect issues early



Quality assurance

Automates testing, analyzes data, and predicts performance to improve quality and speed



AI in Product Launch and Marketing

AI makes launches smarter by optimizing timing, personalizing outreach, and predicting market demand.

Launch timing

Analyzes trends and behavior to identify the optimal launch windows



Personalized marketing

Automates audience segmentation and content creation to boost engagement



Sales forecasting

Predicts demand and trends to guide resource allocation and strategy

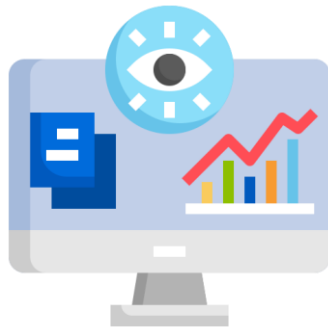


AI in Post-Launch Optimization and Growth

AI drives continuous growth by monitoring product usage, personalizing experiences, and enhancing customer support.

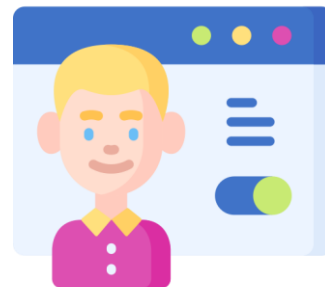
Performance monitoring

Tracks usage and trends to inform continuous product improvements



User personalization

Analyzes behavior and preferences to deliver tailored user experiences



Customer support

Automates responses and gathers feedback to provide live support and issue resolution



Quick Check



How does AI contribute to the product design and development process?

- A. By setting goals and KPIs
- B. By simulating prototypes and detecting issues early
- C. By analyzing launch windows
- D. By automating audience segmentation



Prompt Engineering for Product Lifecycle

Introduction to Prompt Engineering for Product Lifecycle

It is the skill of crafting clear instructions to guide AI models, especially large language models (LLMs), toward desired outputs.

It enables PMs to:

Accelerate creation of user stories, summaries, and other routine tasks

Guide AI to generate high-quality content

Maximize AI impact by learning to steer models effectively

Speed up iteration with quick prototyping and content variations

Understand AI behavior to design smarter, more intuitive features

How Prompts Shape Model Behavior and Output

Prompts act as the steering mechanism for AI models, shaping their behavior, responses, and the quality of outputs they produce.

Effective prompts let PMs:

Guide behavior
Define tone, task, and constraints
to steer AI response

Extract insights
Analyze reviews, feedback, or
market data fast

Generate user stories
Create detailed, goal-aligned
stories instantly

Draft feature copy
Write concise, benefit-driven
product descriptions



Prompt Formats: Zero-Shot, One-Shot, and Few-Shot

Prompt format determines how much guidance the model receives before generating a response.

Zero-shot

The model completes the task using only written instructions, without any examples for reference, making it ideal for simple or familiar tasks.

One-shot

The model follows instructions supported by a single example to guide the response format or tone, especially when consistent style or phrasing is important.

Few-shot

The model learns from multiple examples and applies the pattern to generate more accurate and consistent responses, especially for complex or diverse inputs.

Providing more reference examples helps the model understand the intent and generate more reliable responses.

Prompt Formats in Action

Scenario: You're using an LLM to help write stakeholder update summaries for recent product improvements.
Feature request: *Enable real-time data sync across dashboards for faster insights and improved cross-team visibility.*

Zero-shot

Prompt: *Summarize the feature request for stakeholders.*

Output: *We added real-time data sync to dashboards.*

One-shot

Prompt: *Summarize the feature request like this: We added AI tagging to help users find relevant reports faster.*

Output: *We introduced real-time data sync to help teams access up-to-date insights and improve collaboration.*

Few-shot

Prompt: *Summarize the feature request like this:
We added AI tagging to help users find relevant reports faster.
We streamlined filters to reduce time spent searching for KPIs.*

Output: *We introduced real-time dashboard syncing to boost data accuracy, accelerate decision-making, and improve visibility across teams.*

LLMs detect patterns in the examples provided to improve output quality, forming the core of few-shot prompting.

Fixing Prompt Failures and Debugging Techniques

Prompt failures occur when inputs confuse the model or lack the context it needs to respond accurately.

Common pitfalls that lead to prompt failures:

Ambiguous prompt

Missing context

Conflicting instructions

Poor examples

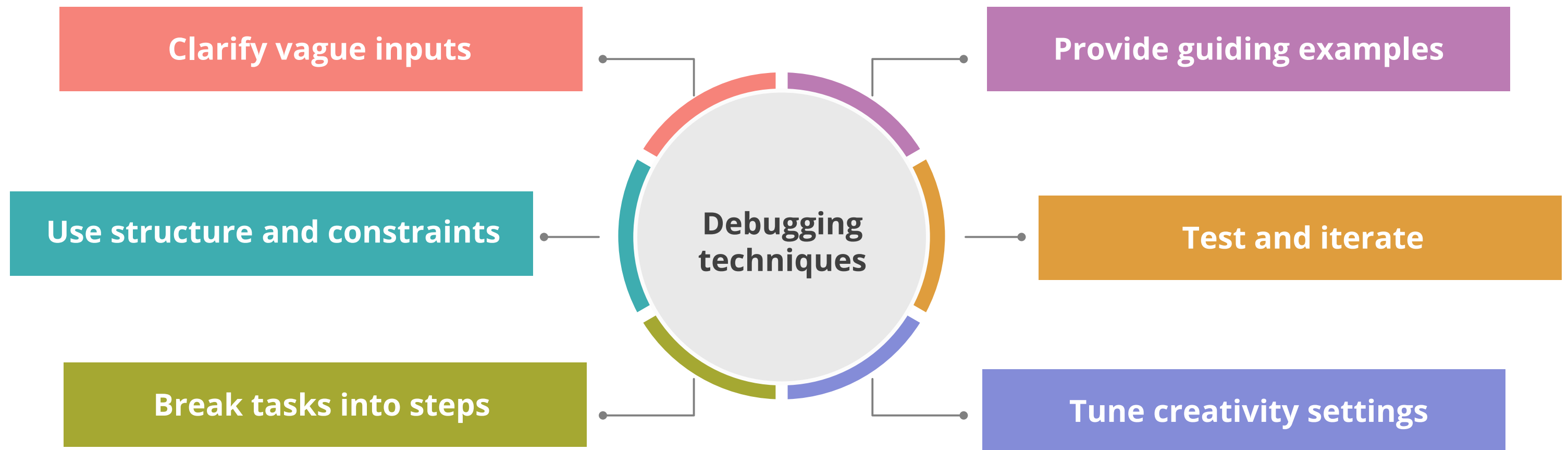
Overly constrained prompt

Model limitations

Most prompt issues arise when inputs are unclear, incomplete, or overly restricted.

Fixing Prompt Failures and Debugging Techniques

Prompt failures from unclear inputs, poor structure, or misaligned tone can be fixed using debugging techniques to improve prompt reliability.



These techniques are especially useful when the model gives hallucinated, generic, or off-topic answers to user questions.

Demo: Refining Prompt Failures in Product Feature Summaries



Duration: 10 minutes

Overview: This demo highlights how vague or poorly structured prompts can result in incorrect or low-quality model responses. It further illustrates how prompt debugging techniques, such as adding context and applying few-shot prompting, can significantly refine responses and improve output quality.

DEMONSTRATION

Quick Check

What is the purpose of few-shot prompting?

- A. To give the model multiple examples to improve consistency
- B. To test the model's response without guidance
- C. To remove constraints and increase creativity
- D. To assign a persona for improved tone control



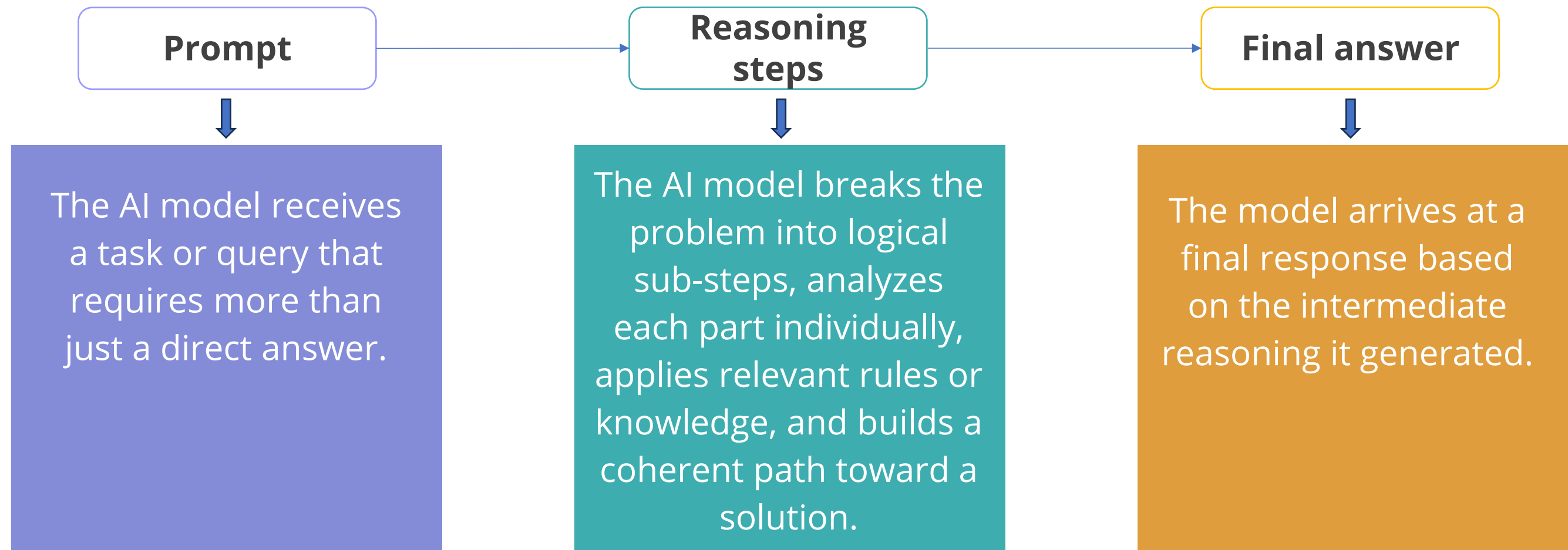


Chain-of-Thought (CoT) Prompting and the ReAct Framework

Chain-of-Thought (CoT) Prompting: Overview

It is a method that guides AI models to break down their thought process by generating intermediate steps between the input and final output, which helps improve reasoning quality and transparency.

Working process of CoT:



How CoT Prompting Supports Decision-Making for PMs

CoT helps product managers generate structured outputs for clearer prioritization and decision-making.

Evaluating feature trade-offs:

It lays out pros and cons in a logical sequence to clarify decision paths.

Creating prioritization matrices:

It applies consistent criteria step by step, improving prioritization accuracy.

Drafting decision summaries:

It produces structured, easy-to-communicate justifications for stakeholder alignment.

Chain-of-Thought Prompting: Use Case

Prompt: *Which feature should we prioritize: real-time notifications or improved onboarding flow?*



LLM model (ChatGPT) output:

Model Output (Chain-of-Thought Reasoning):

Let's break this down:

1. "User Impact: Onboarding affects all new users and helps reduce early churn. Real-time notifications mainly help returning users."
2. "Current Metrics: We're seeing high sign-up rates but low activation — indicating onboarding is a bottleneck."
3. "Team Effort: Improving onboarding is a moderate effort task. Real-time notifications would need backend infra and cross-functional coordination."
4. "Strategic Goal: Increasing early activation and retention is our current focus."

✅ Final Decision: Prioritize improved onboarding flow

Quick Check

How does Chain-of-Thought (CoT) prompting help AI models produce better answers?

- A. By increasing the size of the training dataset
- B. By asking the model to list multiple outcomes
- C. By helping the model think through each step before answering
- D. By removing the need for logical steps in decision-making



ReAct Framework: Overview

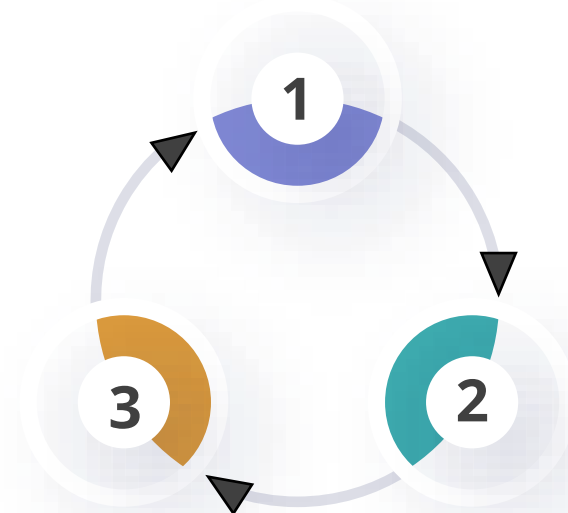
ReAct (Reasoning + Action) enables AI agents to combine CoT reasoning with tool use and outcome-based adaptation, making it well-suited for tasks that require both logical thinking and real-world execution.

Reason:

The agent uses CoT prompting to think through the task step by step and form a logical approach.

Observe:

The agent evaluates outcomes and updates its reasoning for the next step.



Act:

Based on its reasoning, the agent interacts with external tools, such as APIs, search engines, or databases.

This loop continues until the task is complete.

ReAct Framework in PM Tasks

ReAct enables AI agents to fetch real-time data, automate workflows, make data-driven decisions, and adapt to changing user inputs in PM tasks.

Dynamic information retrieval

Handles questions beyond training data by letting the agent use tools like search APIs to fetch real-time data

Multi-step task automation

Solves complex workflows by reasoning through subtasks and invoking tools at each step as needed

Data-driven decisions

Supports informed outputs by integrating live metrics and querying APIs mid-reasoning

Dynamic user interactions

Responds intelligently to user input by adjusting reasoning and actions based on changing queries or tool outputs

ReAct in Action: AI Market Research Example

Prompt: *What are users saying about AppX?*



Step 1:

- **Thought:** I need to analyze what users are saying about AppX.
- **Action:** Search online for AppX reviews across app stores and review sites
- **Observation:** Found multiple review pages with user comments and ratings



Step 2:

- **Thought:** I should extract key points from the reviews.
- **Action:** Identify common likes, complaints, and patterns
- **Observation:** Many users like the design but mention frequent app crashes.



Step 3:

- **Thought:** I need to deliver the main takeaways clearly.
- **Action:** Create a concise summary of user sentiment
- **Observation:** Summary ready to present



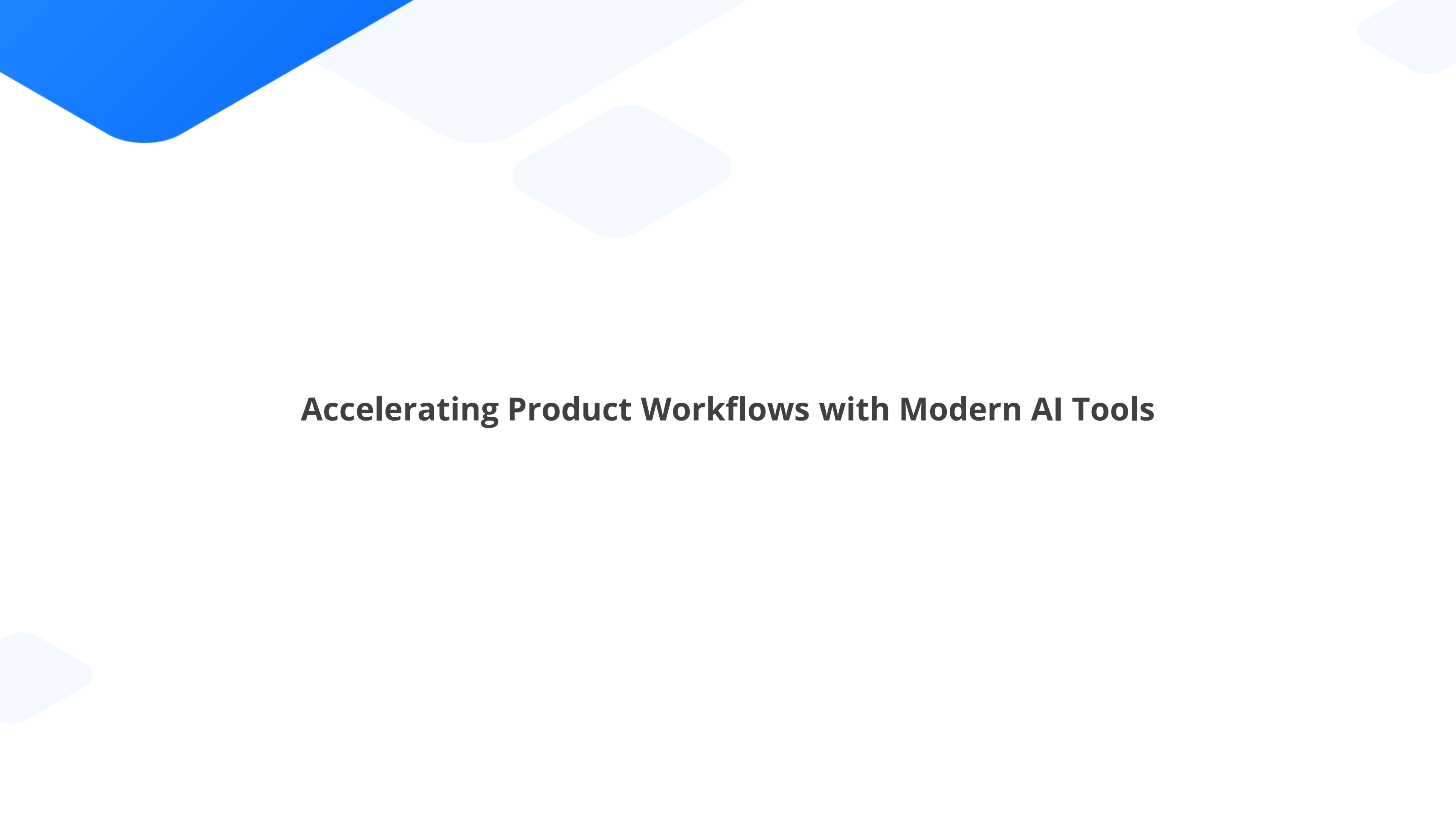
Output: Positive design feedback; app reliability issues flagged

Quick Check



Which statement best explains how ReAct differs from Chain-of-Thought prompting?

- A. ReAct uses tools and APIs to act on the world, while CoT only reasons internally.
- B. ReAct relies only on few-shot examples, unlike CoT.
- C. ReAct is slower but produces more creative outputs than CoT.
- D. ReAct only works on static tasks, while CoT handles dynamic inputs.



Accelerating Product Workflows with Modern AI Tools

AI Tools for Content Generation and Summarization

AI-powered tools help PMs draft content, summarize meetings, and generate ideas, boosting productivity across documentation, communication, and planning.

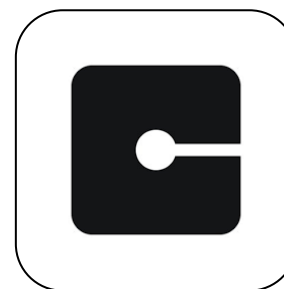
LLMs tools



ChatGPT



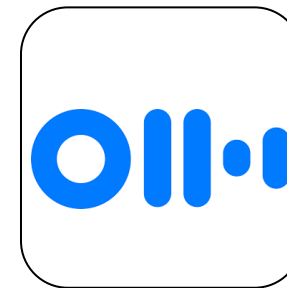
Jasper



Copy.ai

- ➡ Draft UX or marketing copy
- ➡ Summarize documents
- ➡ Brainstorm messaging or product ideas
- ➡ Generate PRD sections or feature outlines

Meeting recording tools



Otter.ai



Fathom

- ➡ Transcribe meetings
- ➡ Capture decisions
- ➡ Auto-generate action items

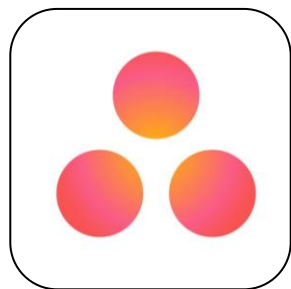
AI Tools for Project Management and Collaboration

AI streamlines PM workflows by automating routine coordination tasks and enhancing access to institutional knowledge across the team.

AI-enhanced PM tools



Jira



Asana

- ➡ Automate task assignments
- ➡ Predict delays
- ➡ Suggest next steps

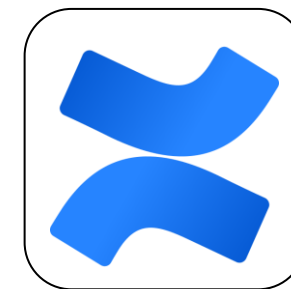
Knowledge management tools



Notion AI



Guru



Confluence

- ➡ Organize internal documentation
- ➡ Improve searchability and access to answers
- ➡ Summarize meeting notes and decisions automatically

Use Case: Using ChatGPT to Draft a PRD

Prompt: *Write a Product Requirements Document (PRD) for a dark mode feature on a mobile app. Include problem statement, user stories, requirements, and success metrics.*

Output:

Product Requirements Document (PRD)

Feature Name: Dark Mode Support for Mobile App

Problem Statement:

Users report discomfort using the app at night due to bright screens. A dark mode option will enhance usability in low-light environments and reduce battery consumption on OLED devices.

User Stories:

- As a user, I want to switch to dark mode at night so I can reduce eye strain.
- As a user, I want the app to remember my dark mode preference across sessions.
- As a user, I want the dark mode to align with my phone's system settings automatically.

Functional Requirements:

- Add toggle in settings to enable/disable dark mode
- Sync with system-level dark/light theme
- Store user preference locally

Success Metrics:

- 80% of users enable dark mode within 30 days
- 20% reduction in support queries related to screen brightness
- 10% increase in user satisfaction scores for nighttime usage

Rapid Prototyping with AI Tools

Early design work often slows progress, but rapid prototyping tools help PMs move faster with less reliance on design teams.

Lovable

Generates full-stack apps with AI-driven front-end and back-end code creation



Uizard

Converts text, sketches, or screenshots into editable UI designs for rapid prototyping

These tools let PMs:

Generate early
UI layouts

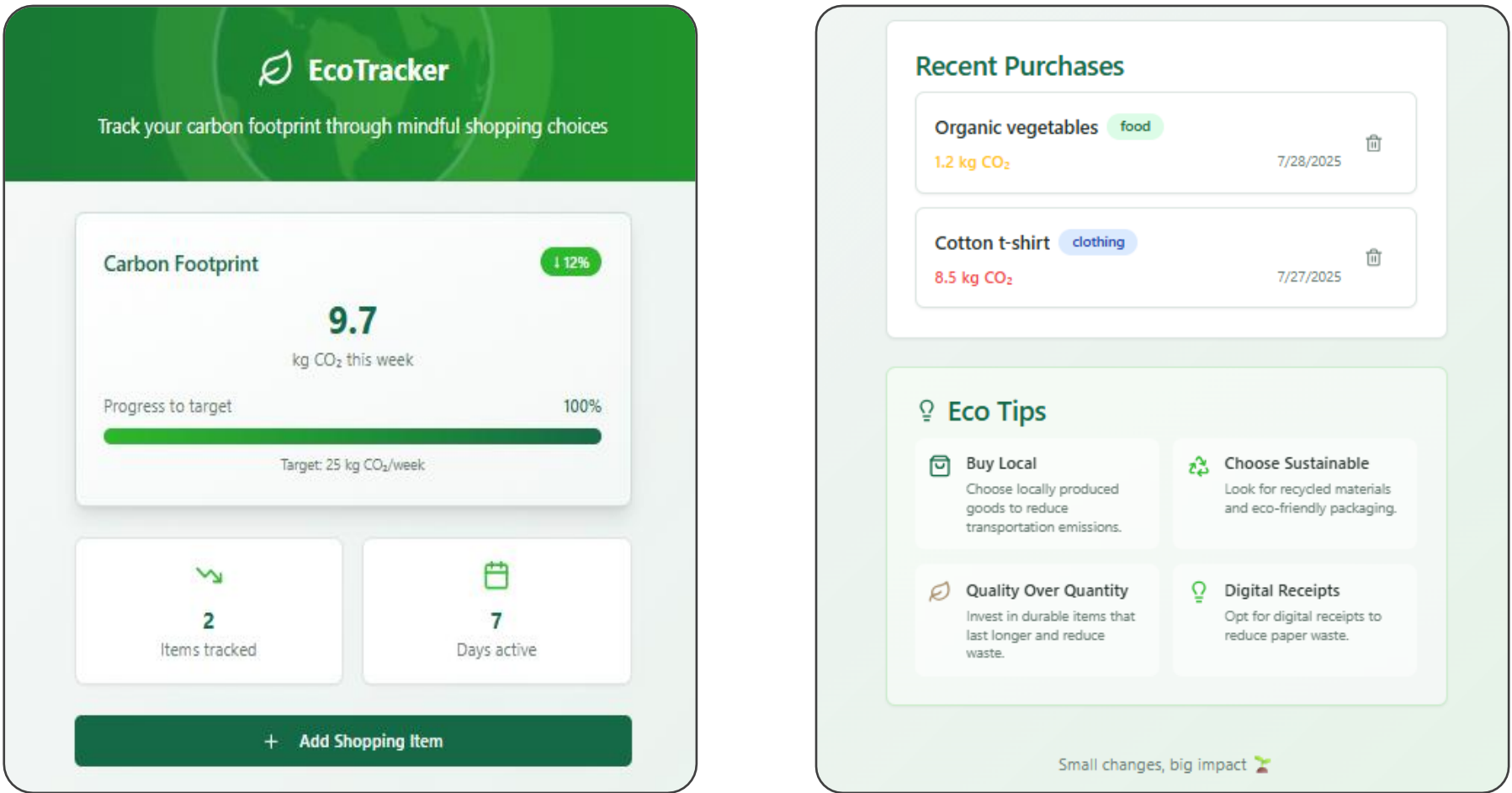
Validate design ideas with
stakeholders

Use Case: Rapid UI Prototyping with Lovable

Scenario: A client requests to preview the UI for a sustainability app that tracks carbon footprint from shopping data. You're short on time and don't have access to design or development resources.

Prompt: *UI for a mobile app that tracks carbon footprint based on shopping details*

Output:



Prototyping App Logic and Agent Workflows with AI Tools

Logic flows and backend behaviors can be prototyped early using no-code tools built for product experimentation.

Firestore

Simulates app logic and user flows using event triggers without needing backend code



Emergent labs

Designs and tests agent workflows for bots and assistants through no-code automation

Key uses include:

Prototyping user logic before development

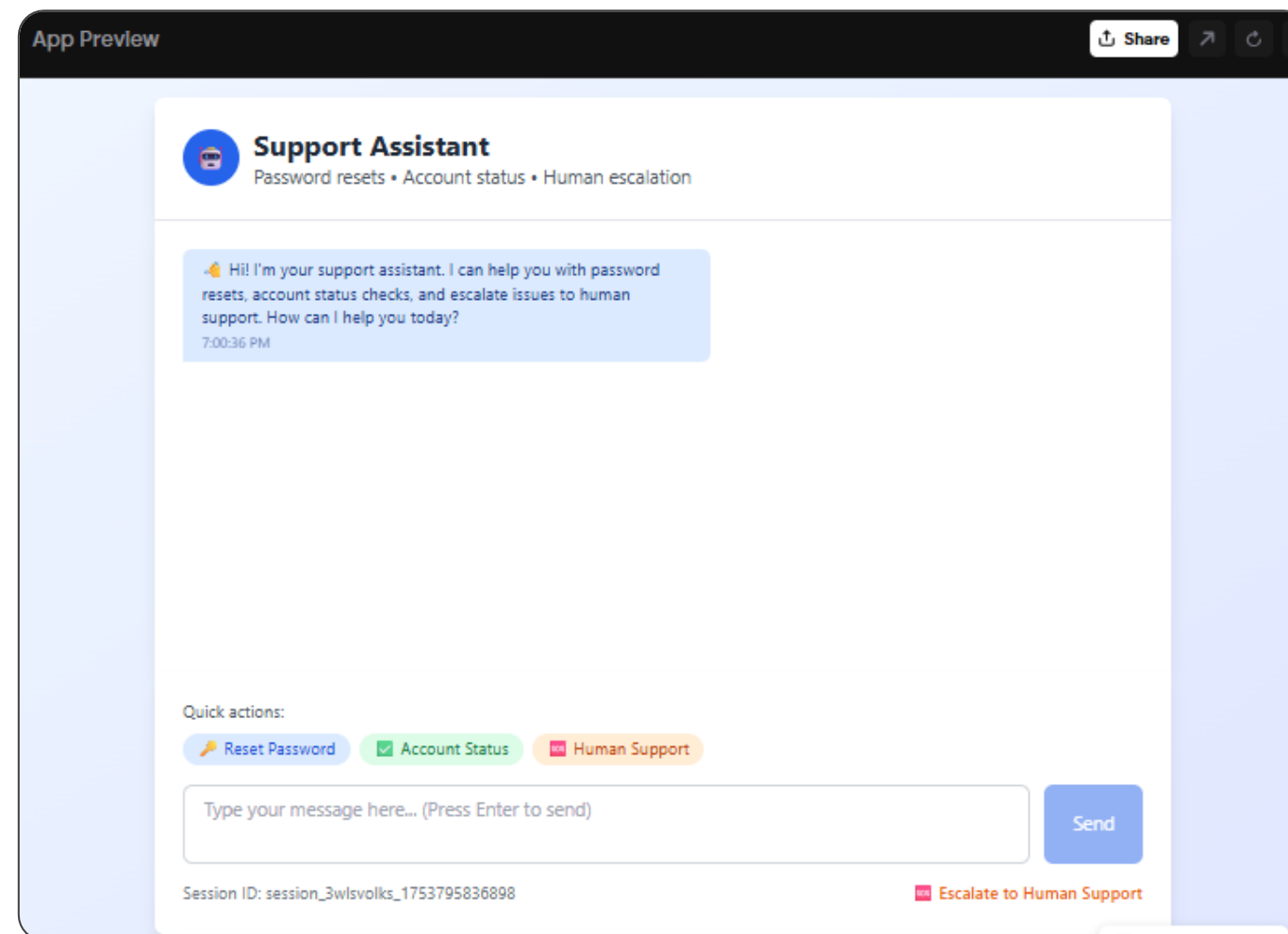
Designing intelligent agent flows for automation

Use Case: Designing an AI Assistant Flow with Emergent Labs

Scenario: You need to launch a support chatbot that can reset passwords, check account status, and escalate issues, without relying on engineering bandwidth or waiting on backend logic to be built.

Prompt: *Design a support chatbot that helps users reset passwords, check account status, and escalate unresolved issues.*

Output:



Quick Check



Which of the following is a benefit of using AI-powered tools like Notion AI or Guru for product teams?

- A. Assigning development tasks to engineers
- B. Predicting delays for key milestones
- C. Organizing documentation and improving knowledge access
- D. Creating front-end interfaces and UI layouts

Demo: Applying CoT and ReAct Prompting for Drafting User Stories and Analyzing Problems



Duration: 10 minutes

Overview: This demo showcases how Chain-of-Thought (CoT) and ReAct prompting techniques enable product managers to reason effectively through user stories and product problems. It also demonstrates how these methods support structured decision-making and tool-based action planning, enhancing clarity and problem-solving efficiency.

Note:

Please download the demo document from the Reference Material section to perform step-by-step execution.

DEMONSTRATION

Guided Practice



Overview

Duration: 15 minutes

You are a product manager exploring how AI-powered tools can support different stages of product development. Your goal is to use a combination of AI tools, Lovable, Uizard, and Firebase Studio to ideate, validate, and simulate a product idea. You'll begin by crafting a strong value proposition, then translate it into a functional user interface prototype, and finally simulate backend functionality to demonstrate how data would flow in a real application. Follow the guided steps to complete each part of this end-to-end product workflow using AI, even without deep technical or design expertise.

GUIDED PRACTICE

Key Takeaways

- AI enhances every stage of product management—from market research to design, launch, iteration, and growth.
- Prompt engineering is a critical skill for PMs to guide AI outputs with clarity, structure, and intent.
- Advanced techniques like Chain-of-Thought and ReAct improve reasoning, planning, and decision-making with AI.
- LLMs like ChatGPT help PMs accelerate tasks such as documentation, brainstorming, user research, and competitive analysis.
- AI-native product management requires PMs to work alongside ML teams, define tool boundaries, and monitor agent behavior.



Q&A

